NMR Spectroscopy

Environments, shift, integration and splitting

Name _____ Class _____ Date _____

^{1}H NMR DATA

δ (ppm): alkyl 0.9–1.7 · next to C=O 2.0–2.7 · on C–O 3.3–4.3 · alkene 4.5–6.5 · aromatic 6.5–8.0 · CHO 9–10 · COOH 10–12 · OH 1–5

δ (ppm) = shift from TMS (Hz) ÷ spectrometer frequency (MHz) · n + 1 rule: n neighbouring H give n + 1 peaks

^{13}C (ppm): alkyl 0–50 · C–O 50–90 · alkene/aromatic 110–160 · C=O 160–220 · IR: C=O 1680–1750 cm^{-1}

Draw the structure and label each H environment. Show units in every calculation step.

A. Environments

1. List the four kinds of information a ^{1}H NMR spectrum provides.

2. Give the number of ^{1}H NMR signals and the ratio of their areas for: (a) ethane (b) propane (c) 2-methylpropane, $(CH_3)_3CH$ (d) benzene (e) methyl propanoate, $CH_3CH_2COOCH_3$

B. Chemical shift

3. (a) A signal is 2220 Hz from TMS on a 600 MHz spectrometer. Calculate δ. (b) How many Hz from TMS is a signal at 1.20 ppm on a 300 MHz spectrometer?

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C. Splitting and predicting spectra

4. Predict the splitting of each signal in: (a) chloroethane, CH_3CH_2Cl (b) 2-bromopropane, $(CH_3)_2CHBr$

5. Predict the full ^1H NMR spectrum (δ, area, splitting) of propanoic acid, CH_3CH_2COOH.

6. How many ^{13}C NMR signals do these give: (a) propanone (b) ethyl ethanoate (c) benzene?

D. Solving structures

7. A compound C_3H_8O shows: doublet (6H) at 1.2 ppm, septet (1H) at 4.0 ppm, singlet (1H) at 2.2 ppm. Identify it and assign each signal.

E. Challenge

8. A compound C_4H_8O has a strong IR absorption at 1715 cm^{-1} and a molecular ion at m/z 72. Its ^1H NMR spectrum shows: triplet (3H) at 1.1 ppm, singlet (3H) at 2.1 ppm, quartet (2H) at 2.4 ppm. Identify the compound and explain each piece of evidence.

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